

London School of Economics
EC400 Mathematics Solutions
September 2011

Not for MSc Finance and Economics Students
Time: 3 hours

This examination has three parts: Revision Mathematics, Mathematics for Microeconomics, and Mathematics for Macroeconomics. Each part is worth 100 marks. Write your name, your class group number, and the name of your class teacher on your script. Explain your answers to get high marks.

Part 1. Revision Mathematics. Answer all questions.

Some of these questions have yes no answers, or numerical answers. You must explain your answers to these questions.

Part 2. Mathematics for Microeconomics. Answer all questions.

5. (20 marks) Consider the following quadratic form:

$$Q(\mathbf{x}) = ax_1^2 + bx_2^2 + 2cx_1x_3$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the leading principal submatrices of A .

Solutions: The matrix of coefficients is:

$$A = \begin{pmatrix} a & 0 & c \\ 0 & b & 0 \\ c & 0 & 0 \end{pmatrix}$$

Its leading principal submatrices are:

$$A_3 = \begin{pmatrix} a & 0 & c \\ 0 & b & 0 \\ c & 0 & 0 \end{pmatrix} \quad A_2 = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \quad A_1 = (a)$$

- (b) (8 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *negative semi-definite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is negative semi-definite?

Solutions: A quadratic form is negative semi-definite if $Q(\mathbf{x}) \leq 0$ for any $\mathbf{x} \neq \mathbf{0}$. The quadratic form is negative semi-definite if all the principal minors of A are of odd size are non-positive, and all the principal minors of A are of even size are non-negative.

- (c) (8 marks) For what values of the parameters a, b, c , is the quadratic form $Q(\mathbf{x})$ *positive definite*? For what values is it *negative definite*? For what values is it *positive semi-definite*?

Solutions: $Q(\mathbf{x})$ is positive definite if and only if

$$|A_1| = a > 0, \quad |A_2| = ab > 0, \quad |A_3| = -bc^2 > 0.$$

but the three conditions cannot hold at once, as $b > 0$ for $|A_2| > 0$ to hold and $b < 0$ for $|A_3| > 0$ to hold. $Q(\mathbf{x})$ is negative definite if and only if

$$|A_1| = a < 0, \quad |A_2| = ab > 0, \quad |A_3| = -bc^2 < 0.$$

but the three conditions cannot hold at once, as $b < 0$ for $|A_2| > 0$ to hold and $b > 0$ for $|A_3| < 0$ to hold. $Q(\mathbf{x})$ is positive semi-definite if and only if all principal minors are non-negative, which amounts to

$$a \geq 0, \quad b \geq 0 \quad \text{and} \quad c = 0.$$

6. (40 marks) For some constant $k > 0$ consider the function defined on $\mathbb{R}^2$:

$$f(x, y) = 2ky(x + 1) - k^2(x^4 + 2x^2)/2 - y^2.$$

- (a) (8 marks) Find the *critical points* of $f(x, y)$.

Solutions: The critical points of function $f(x, y)$ satisfy

$$x^* = 1 \quad \& \quad y^* = 2k.$$

and are the solutions to FOC

$$\frac{df(x, y)}{dx} = 2ky - 2k^2(x^3 + x) = 0 \quad (1)$$

$$\frac{df(x, y)}{dy} = 2k(x + 1) - 2y = 0 \quad (2)$$

- (b) (8 marks) Are these critical points *local maximizers*, *local minimizers* or *neither*?

Solutions: The critical points (x^*, y^*) of the function $f(x, y)$ is a local maximizers since the sufficient second order condition for a local maximizer is satisfied. The hessian satisfies:

$$H = \frac{d^2 f(x, y)}{d(x, y)^2} = \begin{bmatrix} -2k^2(3x^2 + 1) & 2k \\ 2k & -2 \end{bmatrix} \quad (3)$$

Note that it is negative definite as leading principal minors alternate in signs with the appropriate pattern at the critical point, $H_1 = -2k^2(3x^2 + 1) < 0$ and $H_2 = 12k^2x^2 > 0$.

- (c) (8 marks) Are these critical points *global maximizers*, *global minimizers* or *neither*?

Solutions: The critical point (x^*, y^*) is also a global maximizer of the function $f(x, y)$ since the function is globally concave the Hessian is negative semi-definite for any (x, y) .

- (d) (8 marks) Is $f(x, y)$ *concave*, *convex*, *quasi-concave*, *quasi-convex* or *neither of the above*?

Solutions: The concavity of function follows from the solution to part c. Thus, the function is also quasi-concave. The function is clearly not convex as it is strictly concave on parts of the domain (e.g. around the optimum). Finally the function is not quasi-convex. Consider the points $(1, -1)$ and $(-1, 1)$ and their convex combination $1/2(1, -1) + 1/2(-1, 1) = (0, 0)$, and observe that $f(-1, 1) = -3k^2/2 - 1$, $f(1, -1) = -4k - 3k^2/2 - 1$, but $f(0, 0) = 0$. Thus the function is not quasi-convex as

$$f(0, 0) > \max\{f(1, -1), f(-1, 1)\}.$$

- (e) (8 marks) Denote the *global maximizer* of $f(x, y)$ by $(x(k), y(k))$. Use the envelope theorem to show how *maximum value function* responds to a marginal change in k . Also show how the *global maximizer* $(x(k), y(k))$ is affected by marginal changes in k .

Solutions: By the envelope theorem we know that:

$$\begin{aligned}\frac{d}{dk}f(x(k), y(k), k) &= \frac{\partial}{\partial k}f(x(k), y(k), k) = \\ &= 2y(k)(x(k) + 1) - k(x(k)^4 + 2x(k)^2) \\ &= 5k > 0\end{aligned}$$

Furthermore since $x(k) = 1$ and $y(k) = 2k$:

$$x'(k) = 0 \text{ and } y'(k) = 2 > 0$$

7. (40 marks) For $p > 0$, consider the problem:

$$\begin{aligned}\max_{x,y} \quad & F(x, y) = x^{3/4}y^{1/4} \\ \text{s.t.} \quad & x + py \leq 10 \\ & x \geq 0 \\ & y \geq 0\end{aligned}\tag{4}$$

- (a) (8 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Which conditions on the concavity/convexity of such functions would suffice to guarantee that the Lagrangian is concave?

Solutions: The objective function is concave since its Hessian is negative semi-definite in $x, y \geq 0$:

$$\begin{aligned}d^2F(x, y) &= \frac{3}{16} \begin{pmatrix} -x^{-5/4}y^{1/4} & x^{-1/4}y^{-3/4} \\ x^{-1/4}y^{-3/4} & -x^{3/4}y^{-7/4} \end{pmatrix} \\ \text{since } |d^2F(x, y)| &= 0, \quad -x^{-5/4}y^{1/4} \leq 0 \quad \text{and} \quad -x^{3/4}y^{-7/4} \leq 0\end{aligned}$$

The constraints are obviously linear (both convex and concave). Because the objective function is concave and the constraints are convex, the Lagrangian is concave since it is a sum of concave functions.

- (b) (8 marks) Set up the constrained optimization problem using the Lagrangian function. Can we solve this optimization problem using the Kuhn-Tucker method? Are the Kuhn-Tucker constraint qualifications satisfied? Is it possible to solve the same optimization problem with the Lagrangean method? Are the Lagrange constraint qualifications satisfied?

Solutions: The Lagrangian of the problem is

$$\mathcal{L}(x, y, \lambda_1, \lambda_2, \lambda_3) = x^{3/4}y^{1/4} - \lambda_1(x + py - 10) + \lambda_2 x + \lambda_3 y$$

Notice first that a simple contradiction argument proves that if $p < \infty$ the optimal solution cannot have $x = 0$ and/or $y = 0$, as we could increase the value of the program by setting $x = 9$ and $y = 1/p$.

The Lagrangian of the problem then becomes:

$$\mathcal{L}(x, y, \lambda_1, \lambda_2, \lambda_3) = x^{3/4}y^{1/4} - \lambda_1(x + py - 10)$$

The constraint set has a non-empty interior thus we can adopt the Kuhn-Tucker approach to solve the problem. Moreover, the gradient of the constraint $g(x, y) = x + py - 10$ is:

$$\nabla g(x, y) = (1, p)$$

therefore constraint qualifications for the Lagrange method are satisfied for every pair (x, y) such that $x + 2p \leq 10$. In other words, we can use the Lagrangian method to solve the constrained maximization problem.

- (c) (8 marks) Write down the necessary first order conditions for the solution of the constrained optimization problem.

Solutions: The necessary first order conditions for the solution of the constrained maximization problem are then:

$$\begin{aligned} 0 &= (3/4)(y/x)^{1/4} - \lambda_1 \\ 0 &= (1/4)(x/y)^{3/4} - p\lambda_1 \\ 0 &= \lambda_1(x + py - 10) \\ 0 &\geq x + py - 10 \\ 0 &\leq \lambda_1 \end{aligned}$$

- (d) (4 marks) Are these conditions also *sufficient* to characterize an optimum in this problem?

Solutions: The necessary first order conditions for the solution of the constrained maximization problem are sufficient for a maximum since the objective function $f(x, y)$ is concave.

- (e) (4 marks) Solve the constrained optimization problem in terms of p .

Solutions: The solution to the constraint optimization problem is the solution to the necessary and sufficient conditions above and is $\lambda_1 = (3/4)(1/3p)^{1/4}$ and $(x^*(p), y^*(p)) = (15/2, 5/2p)$.

- (f) (8 marks) Let $x^*(p)$ and $y^*(p)$ be the solution of this problem. Find the slope of such functions with respect to p . Moreover, find how the maximum value function is affected by a change in p .

Solutions: Since the bordered Hessian is negative definite at the global optimum, the implicit function theorem applies. Thus, it is immediate to observe that:

$$\frac{\partial}{\partial p}x^*(p) = 0 \text{ and } \frac{\partial}{\partial p}y^*(p) = -\frac{5}{2p^2}.$$

Finally, observe that by the Envelope theorem:

$$\begin{aligned} \frac{d}{dp}F(x^*(p), y^*(p)) &= \\ &= \frac{\partial}{\partial x}F(x^*(p), y^*(p))\frac{\partial}{\partial p}x^*(p) + \frac{\partial}{\partial p}F(x^*(p), y^*(p))\frac{\partial}{\partial p}y^*(p) = \\ &= \lambda_1 \left[\frac{\partial}{\partial x}g(x^*(p), y^*(p), p)\frac{\partial}{\partial p}x^*(p) + \frac{\partial}{\partial y}g(x^*(p), y^*(p), p)\frac{\partial}{\partial p}y^*(p) \right] = \\ &= -\lambda_1 y^*(p) = -(15/8p)(1/3p)^{1/4} \end{aligned}$$

for $g(x, y, I) = x + py - 10$.

Part 3. Mathematics for Macroeconomics. Answer all questions.

8. **(50 marks)** Let $\alpha \in (0, 1)$, $\delta \in (0, 1)$, $\rho \in (0, 1)$, and $\sigma > 0$. Impose that $\rho + \delta < 1$. Given $k(0)$, we want to find the function $c(t)$ to maximize

$$\int_0^{+\infty} e^{-\rho \cdot t} \cdot \frac{c(t)^{1-\sigma} - 1}{1 - \sigma} dt,$$

subject to the law of motion

$$\dot{k}(t) = k(t)^\alpha - c(t) - \delta \cdot k(t).$$

- (a) (20 marks) Which variable do you choose as a state variable? Which variable do you choose as a control variable? Write down the current-value Hamiltonian and derive the optimality conditions.

SOLUTION: (5 marks) State variable: $k(t)$. Control variable: $c(t)$. Current-value Hamiltonian:

$$H^*(t) = \frac{c(t)^{1-\sigma} - 1}{1 - \sigma} + q(t) \cdot [k(t)^\alpha - c(t) - \delta \cdot k(t)],$$

where $q(t)$ is the co-state variable.

(5 marks) First optimality condition:

$$\begin{aligned} \frac{\partial H^*(t)}{\partial c(t)} &= 0 \\ c(t)^{-\sigma} &= q(t). \end{aligned}$$

(5 marks) Second optimality condition:

$$\begin{aligned} \frac{\partial H^*(t)}{\partial k(t)} &= \rho \cdot q(t) - \dot{q}(t) \\ q(t) \cdot [\alpha \cdot k(t)^{\alpha-1} - \delta - \rho] &= -\dot{q}(t) \\ [\alpha \cdot k(t)^{\alpha-1} - \delta - \rho] &= -\frac{\dot{q}(t)}{q(t)} \end{aligned}$$

(5 marks) Third optimality condition: the transversality condition

$$\lim_{t \rightarrow +\infty} e^{-\rho \cdot t} \cdot q(t) \cdot k(t) = 0.$$

- (b) (5 marks) The Euler equation is the first-order differential equation that characterizes the optimal function $c(t)$. Determine the Euler equation.

SOLUTION: Take the log of the first optimality condition, differentiate with respect to time t , and plug the result from the second optimality condition:

$$\begin{aligned} -\sigma \cdot \ln[c(t)] &= \ln[q(t)] \\ \sigma \cdot \frac{\dot{c}(t)}{c(t)} &= -\frac{\dot{q}(t)}{q(t)} \\ \frac{\dot{c}(t)}{c(t)} &= \frac{1}{\sigma} \cdot [\alpha \cdot k(t)^{\alpha-1} - \delta - \rho]. \end{aligned}$$

- (c) (10 marks) Suppose $\alpha = 1$ and $\sigma = 1$. Show that the system describing the optimal functions $\{k(t), c(t)\}$ reduces to a linear, homogenous system of first-order differential equations. Show that the system is unstable by computing the eigenvalues.

SOLUTION: (5 marks) The optimal functions $\{k(t), c(t)\}$ are described by a system of two equations: the Euler equation and the law of motion of $k(t)$. If $\alpha = 1$ and $\sigma = 1$, the system is

$$\begin{bmatrix} \dot{c}(t) \\ \dot{k}(t) \end{bmatrix} = \begin{bmatrix} (1 - \rho - \delta) & 0 \\ -1 & (1 - \delta) \end{bmatrix} \begin{bmatrix} c(t) \\ k(t) \end{bmatrix}.$$

This is a linear, homogenous system of first-order differential equations.

(5 marks) We compute the eigenvalues λ of the system. λ solves

$$\det \begin{bmatrix} 1 - \rho - \delta - \lambda & 0 \\ -1 & 1 - \delta - \lambda \end{bmatrix} = 0.$$

Hence, λ solves

$$[(1 - \rho - \delta) - \lambda] \cdot [(1 - \delta) - \lambda] = 0.$$

Therefore the system admits two distinct positive eigenvalues:

$$\begin{aligned} \lambda_1 &= 1 - (\rho + \delta) > 0 \\ \lambda_2 &= 1 - \delta > 0. \end{aligned}$$

We are facing a linear homogenous system of first-order differential equations with two positive eigenvalues are positive: the system is unstable.

- (d) (15 marks) Suppose $\alpha < 1$ and $\sigma > 0$. Show that the system describing the optimal functions $\{k(t), c(t)\}$ reduces to a nonlinear system of first-order differential equations. Use a phase-diagram to show that the steady state of the system is a saddle point. Explain how you draw the phase diagram.

SOLUTION: (5 marks) The optimal functions $\{k(t), c(t)\}$ are described by a system of two equations: the Euler equation and the law of motion of $k(t)$. If $\alpha < 1$, the system is

$$\begin{aligned} \dot{c}(t) &= \frac{1}{\sigma} \cdot [\alpha \cdot k(t)^{\alpha-1} - \delta - \rho] \cdot c(t) \\ \dot{k}(t) &= k(t)^\alpha - c(t) - \delta \cdot k(t). \end{aligned}$$

This is a nonlinear system of first-order differential equations.

(5 marks) We draw the phase diagram with the state variable k on the x-axis and the control variable c on the y-axis. The locus $\dot{c}(t) = 0$ and the locus $\dot{k}(t) = 0$ satisfy

$$\begin{aligned} k(t) &= \left[\frac{\delta + \rho}{\alpha} \right]^{-1/(1-\alpha)} \equiv k^* \\ c(t) &= k(t)^\alpha - \delta \cdot k(t). \end{aligned}$$

The locus $\dot{c}(t) = 0$ is a vertical line. The locus $\dot{k}(t) = 0$ is a concave curve that goes through the origin and that cuts the x-axis again at $k^{**} = \delta^{-1/(1-\alpha)}$. Since $\rho > 0$ and $\alpha < 1$, $k^{**} > k^*$ and the concave curve crosses the vertical curve when it is positive. The steady state of the system is the intersection of the locus $\dot{c}(t) = 0$ and the locus $\dot{k}(t) = 0$.

(5 marks) Look at the equation for $\dot{c}(t)$ to determine the vertical arrows. Since $\alpha \cdot k^{\alpha-1} - \delta - \rho$ decreases with k , $\dot{c}(t) > 0$ to the west of $\dot{c}(t) = 0$ and $\dot{c}(t) < 0$ to the east of $\dot{c}(t) = 0$. So the vertical arrows point northwards to the west of $\dot{c}(t) = 0$ and southwards to the east of $\dot{c}(t) = 0$.

Look at the equation for $\dot{k}(t)$ to determine the horizontal arrows. Clearly, $\dot{k}(t) > 0$ to the south of $\dot{k}(t) = 0$ and $\dot{k}(t) < 0$ to the north of $\dot{k}(t) = 0$. So the horizontal arrows point westwards to the north of $\dot{k}(t) = 0$ and eastwards to the south of $\dot{k}(t) = 0$.

Therefore, the steady state of the system is a saddle point. The saddle path goes through the south-west and north-east regions of the plane.

9. (50 marks) Let $\beta \in (0, 1)$ and $r > 0$. Given $k_0 > 0$, we want to find a collection of sequences $\{c_t, k_{t+1}\}_{t=0}^{+\infty}$ to maximize

$$\sum_{t=0}^{\infty} \beta^t \cdot \ln(c_t),$$

subject to the constraints

$$k_{t+1} = (1 + r) \cdot k_t - c_t$$

for all $t \geq 0$.

- (a) We first solve the maximization problem using the Lagrangian method.

1. (5 marks) Write down the Lagrangian of the problem.

SOLUTION: The Lagrangian associated with the problem is

$$L = \sum_{t=0}^{+\infty} \beta^t \cdot \{\ln(c_t) - \lambda_t \cdot [k_{t+1} - (1 + r) \cdot k_t + c_t]\},$$

where $\{\lambda_t\}_{t=1}^{+\infty}$ is the sequences of Lagrange multipliers associated with the sequences of constraints.

2. (5 marks) Derive the first-order condition(s) of the maximization problem.

SOLUTION: The first-order conditions with respect to c_t and k_{t+1} are

$$\begin{aligned} \frac{1}{c_t} &= \lambda_t \\ \lambda_t &= \beta \cdot (1 + r) \cdot \lambda_{t+1}. \end{aligned}$$

3. (5 marks) Derive the Euler equation.

SOLUTION: Combining the first-order conditions yields the Euler equation:

$$\frac{c_{t+1}}{c_t} = \beta \cdot (1 + r).$$

(b) We now solve the maximization problem using the dynamic programming method.

1. (5 marks) Which variable do you choose as a state variable? Which variable do you choose as a control variable? Write down the Bellman equation.

SOLUTION: State variable: k . Control variable: k' (the value of variable k next period). Bellman equation:

$$V(k) = \max_{k'} [\ln((1+r) \cdot k - k') + \beta \cdot V(k')].$$

2. (5 marks) Derive the first-order condition associated with the Bellman equation.

SOLUTION: The first-order condition with respect to k' in the Bellman equation is

$$\frac{1}{c} = \beta \cdot V'(k').$$

3. (5 marks) Derive the Benveniste-Scheinkman equation.

SOLUTION: We apply the envelope theorem to the Bellman equation:

$$V'(k) = \frac{(1+r)}{c}.$$

4. (5 marks) Derive the Euler equation. Compare it with the Euler equation obtained with the Lagrangian method and discuss.

SOLUTION: The Benveniste-Scheinkman equation holds for any k . In particular, $V'(k') = (1+r)/c'$. Combining this equation with the first-order condition yields the Euler equation:

$$\frac{c'}{c} = \beta \cdot (1+r).$$

This Euler equation is the same as that obtained with the Lagrangian method. The two methods are equivalent.

5. (5 marks) Suppose that the policy function takes the form $h(k) = A \cdot (1+r) \cdot k$ where $A \in (0, 1)$. Derive A .

SOLUTION: We guess that optimal consumption $c = h(k) = A \cdot (1+r) \cdot k$. A first implication is that

$$\frac{c'}{c} = \frac{A \cdot (1+r) \cdot k'}{A \cdot (1+r) \cdot k} = \frac{k'}{k}.$$

Using the Euler equation, we obtain

$$\frac{k'}{k} = \frac{c'}{c} = (1+r) \cdot \beta.$$

The transition equation then implies

$$c = (1+r) \cdot k - k' = (1-\beta) \cdot (1+r) \cdot k.$$

Therefore, it must be that

$$A = (1-\beta).$$

6. (10 marks) Suppose that the value function takes the form $V(k) = B + D \cdot \ln(k)$, where B and D are constants. Using the expression for the policy function that you derived in the previous question, derive B and D .

SOLUTION: The Bellman equation can be written in terms of the policy function:

$$V(k) = \ln(h(k)) + \beta \cdot V((1+r) \cdot k - h(k)).$$

We plug our guess for the value function and the expression for the policy function into the Bellman equation:

$$B + D \cdot \ln(k) = \ln((1-\beta) \cdot (1+r) \cdot k) + \beta \cdot [B + D \cdot \ln(\beta \cdot (1+r) \cdot k)].$$

Rearranging the terms on the right-hand side yields

$$B + D \cdot \ln(k) = [\ln((1-\beta) \cdot (1+r)) + \beta \cdot B + \beta \cdot D \cdot \ln(\beta \cdot (1+r))] + [1 + \beta \cdot D] \cdot \ln(k).$$

This equation must hold for any k so it is necessary that (5 marks)

$$\begin{aligned} D &= 1 + \beta \cdot D \\ D &= \frac{1}{1-\beta} \end{aligned}$$

and (5 marks)

$$\begin{aligned} B &= \ln((1-\beta) \cdot (1+r)) + \beta \cdot B + \beta \cdot D \cdot \ln(\beta \cdot (1+r)) \\ B &= \frac{(1-\beta) \cdot \ln((1-\beta) \cdot (1+r)) + \beta \cdot \ln(\beta \cdot (1+r))}{(1-\beta)^2} \\ B &= \frac{(1-\beta) \cdot \ln(1-\beta) + \beta \cdot \ln(\beta) + \ln(1+r)}{(1-\beta)^2}. \end{aligned}$$